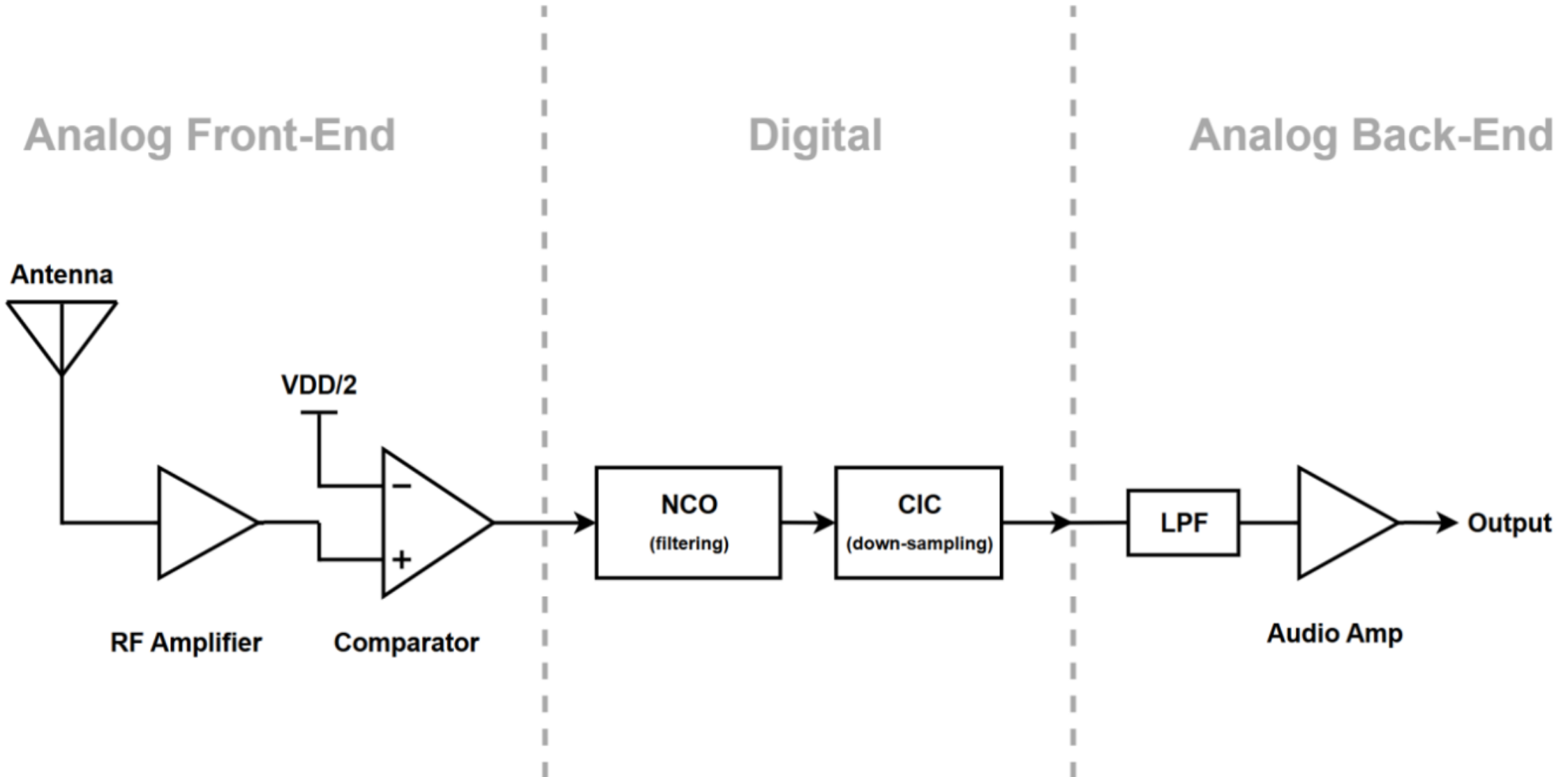


DIGITAL RADIO



DIGITAL RADIO

I/O Pin Assignments

Pin Name	Purpose
clk	Clock (50 MHz)
rst_n	Active-low system reset
ui_in[0]	Digitized 1-bit RF input
ui_in[7:1]	Frequency control
uo_out[0]	PWM output signal
uo_out[1]	NCO debug output

Proposed Specifications

Specification	Value	Notes
Tuning Bandwidth	535kHz – 1800 kHz	Frequency range for AM radio broadcasts
NCO Bit Width	16 bits	Determines how finely we can tune to specific frequencies. 16 bits should give us good precision without taking up too much space
Decimation Factor	1024	Lowers sample rate from 50MHz to ~48kHz, which is standard for audio.
Audio Resolution	~10-bit	Below CD quality but good enough and saves on space

Tiny Tapeout Restrictions

Specification	Value	Notes
Clock Speed	50MHz	Limits the frequencies we can measure (can't receive FM radio signals)
Demo-board Voltage	3.3V	Determines what we read as HIGH (1) or LOW (0)

Timeline of Completion / Task Assignment	
Date	Description of Task / Task Assignment
Week 3 (Sept 24 – Oct 1)	Project Proposal [Both] ☐ Preliminary research ☐ Create block diagram ☐ Define I/O assignments ☐ Define proposed project specifications
Week 4 (Oct 1 – Oct 8)	Verilog code [Both] ☐ Create project Verilog file structure ☐ Assign and initiate I/O wires and paths according to I/O table ☐ Download LTSpice
Week 5 (Oct 8 – Oct 15)	Verilog code ☐ NCO implementation with digital mixer [Angela] ☐ CIC implementation [Ethan] (Reading Week)
Week 6 (Oct 15 – Oct 22)	Verilog code [Angela] ☐ PWM Audio Output implementation ☐ Create CocoTB tests for previously implemented components Analog design [Ethan] ☐ Initial circuit designs in LTSpice

Timeline of Completion / Task Assignment	
Date	Description of Task / Task Assignment
Week 7 (Oct 22 – Oct 29)	(Midterm Week)
Week 8 (Oct 29 – Nov 5)	Analog design ☐ Use LTSpice simulation for testing / verification [Angela] ☐ Tune component specs based on testing [Ethan]
Week 9 (Nov 5 – Nov 12)	Review / Testing [Both] ☐ Run synthesis with TinyTapeout ☐ Debug and review synthesis errors ☐ Work on any areas that require attention
Week 10 (Nov 12 – Nov 19)	Review / Testing [Both] ☐ Run synthesis with TinyTapeout ☐ Debug and review synthesis errors ☐ Work on any areas that require attention
Week 11 (Nov 19 – Nov 26)	Review / Documentation [Both] ☐ Update all documentation on GitHub (/docs folder) ☐ Final preparations for Final Submission